

AeroScale: A Resizable Drone-based POV Display for Mobile and Shared Augmented Reality

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ABSTRACT

Drone-based displays leverage their physical mobility and inherent sharability to ground digital content in the physical world, creating a form of shared augmented reality accessible without individual headsets. However, their potential is fundamentally constrained by a fixed form factor, which enforces a trade-off between mobility (navigational agility) and sharability (display visibility). To this end, this paper presents AeroScale, which features a retractable persistence-of-vision mechanism capable of actively varying its display diameter in flight. This design addresses the trade-off between mobility and sharability, allowing the AR display to prioritize mobility in confined environments or maximize sharability for public conditions.

Index Terms: Augmented Reality, 3D User Interfaces (3D UI), Resizable Display, Mid-air Display, Drone

1 INTRODUCTION

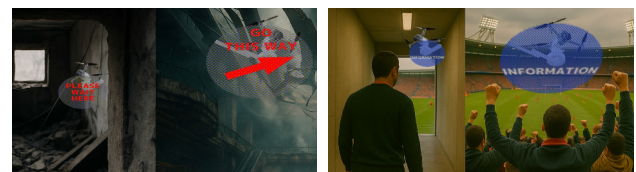
Drones, with their unique ability to freely navigate three-dimensional space, are becoming a compelling platform for creating mid-air 3D user interfaces (UIs) and augmented reality (AR) experiences. Unlike conventional head-mounted displays (HMDs) that achieve AR through optical superposition, such drone-based UIs realize spatial registration by virtue of the drone's physical presence, anchoring digital information to specific real-world coordinates. Given this feature, drone-based UIs bring digital content into our shared physical world with two key advantages over traditional, isolated, user-centric AR views. First, their physical mobility is independent of any single user, allowing them to act as autonomous information agents that can traverse complex environments and reach distant locations. Second, their displays are inherently public, creating a shared AR experience for co-located groups without the need for individual eyewear.

Recognizing this potential, researchers have explored various approaches to creating such drone-based mid-air UIs, each with distinct characteristics. First, formation-based drone displays, exemplified by drone shows [3], use multiple drones equipped with LEDs to create large-scale, three-dimensional visual spectacles. However, managing multiple drones simultaneously carries inherent risks, such as collisions and communication failures. Second, screen-mounted drones [2] project images onto airborne lightweight screens from ground-based projectors. This method offers excellent safety, while their display mobility is limited due to the (often fixed) ground-based projectors. Third, projector-equipped drones [1] use pico-projectors to display images onto building facades or ground-level objects. However, it requires relatively flat projection surfaces and suffers from low daytime visibility due to the limited brightness of the onboard projectors. Fi-

nally, rotational-display drones [4] leverage the POV effect, using rapidly rotating LED arrays to generate visible light-based aerial displays. However, their main limitation lies in the fixed rotation radius, which creates an inherent trade-off between the drone's overall size and the display area—making it difficult for a single drone to both navigate narrow environments and provide large, highly visible displays.

However, existing rotational-display drones, like most other mid-air UIs, suffer from a critical limitation: their physical form factor is fixed. This imposes a static and unavoidable trade-off that directly compromises their core advantages: the conflict is between navigational agility (essential for reaching the AR registration point) and display visibility (essential for shared viewing). A drone with a large display is highly visible for groups but too cumbersome for agile navigation. Conversely, a small drone is mobile but ineffective as a shared display. This conflict prevents any single fixed-size system from fully realizing the potential of mobile and shareable mid-air UIs.

In this paper, we introduce AeroScale, a drone-borne mid-air POV display system featuring a dynamically resizable visualization. Building on existing POV technology, it further incorporates a retractable mechanism that enables real-time variation of the display radius. This resizable design offers a path toward adaptive drone-based AR by allowing the UI to physically conform to the spatial context. For example, in disaster relief (Fig. 1a), AeroScale can prioritize mobility by contracting its radius to navigate tight spaces, then prioritize sharability by expanding to deliver large-scale guidance to a group. In a public venue (Fig. 1b), it can fluidly transition between a personal information proxy (small, mobile) and a public information panel (large, shared).



(a) Disaster relief scenario

(b) Stadium scenario

Figure 1: AR Application Scenarios of AeroScale

2 PROTOTYPE DESIGN AND IMPLEMENTATION

As shown in Fig. 2, the AeroScale prototype consists of three main components mounted on a central fixed frame (3): (i) a drone, our prototype implementation uses the DJI Air 3; (ii) a rotational system for the POV display (1, 2, 4, 6, 7, 8, 9, 10, 12); and (iii) a retractable arm mechanism designed to dynamically adjust the display radius (10, 11, 13, 14, 15, 16). Additionally, the rotational and retractable systems are tightly integrated, sharing key components such as the ESP32 controller (10) and the upper rotating LED array (16).

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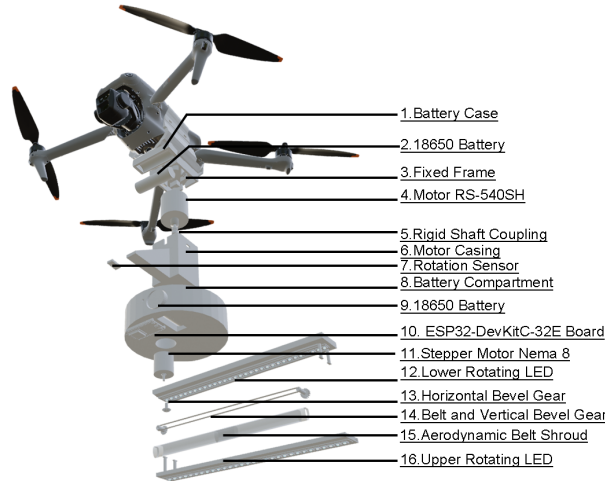


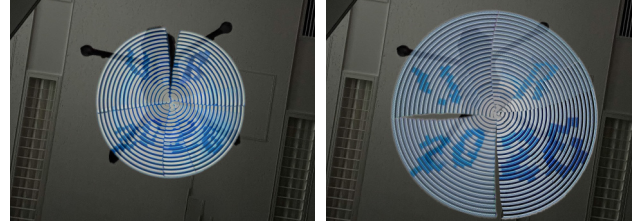
Figure 2: Exploded View of the AeroScale Prototype with Mechanical and Electronic Components Labeled.

Rotational POV Display. POV displays leverage a human visual phenomenon in which rapidly moving lights create the illusion of continuous images suspended mid-air. As detailed in Fig. 2, our implementation is centrally managed by an ESP32-DevKitC-32E board (10). This controller is powered by a single 18650 battery (9) housed in a custom battery compartment (8). Rotational motion is generated by a high-speed RS-540SH motor (4), which is powered by a separate 18650 battery (2) to ensure a stable supply. The motor, enclosed in a motor casing (6), connects to the rotating assembly via a rigid shaft coupling (5), spinning both the lower and upper rotating LED arrays (12, 16). To produce a coherent and stable image, a rotation sensor (7) provides continuous, real-time feedback on rotational speed to the controller. The ESP32 uses this data to dynamically calculate and adjust the timing of LED illumination, ensuring synchronization with the display's current radius (determined by the Nema 8 stepper motor, 11) and rotational speed.

Retractable Arm Mechanism. Our proposed AeroScale distinguishes itself by dynamically adjusting the radius of its POV display through a novel retractable arm mechanism. Figure 2 illustrates the structural details of this system, which is mounted on the drone via a fixed frame (3). Extension and retraction are precisely controlled by a Nema 8 stepper motor (11). When activated, this motor rotates a horizontal bevel gear (13), which engages with a belt looped around a vertical bevel gear (14), effectively translating the motor's rotational input into linear motion. This transmission mechanism smoothly extends or retracts the upper arm segment that holds the upper rotating LED (16), thereby changing the overall display diameter. The entire drive system is enclosed within an aerodynamic belt shroud (15), designed to minimize air resistance and turbulence, ensuring mechanical stability during high-speed rotation and protecting the mechanism from airflow interference generated by the drone's propellers.

Prototype Implementation and Demonstration. Figure 3 presents a 3D-printed AeroScale prototype ($\approx 570\text{g}$), tested in a laboratory environment. It achieves a stable 1500 RPM rotation while hovering and can dynamically adjust the display radius within 10 seconds during operation. This mechanism relies on the lower (12) and upper (16) LED arrays. In the retracted state (Fig. 3a), the lower array is obscured, utilizing only the upper 19 LEDs for a compact 12 cm radius (comparable to a 12.8-inch screen). Upon extension (Fig. 3b), the upper array moves outward to expose the lower one. Activating both arrays increases the count to 26 LEDs, forming a continuous light bar. This expands the radius to 17 cm, yielding a

display area 2.01 times larger. Regarding visual performance, the 1500 RPM rotation ensures a flicker-free display. The flat design removes geometric distortion, though oblique viewing angles lead to perspective shortening due to the downward tilt. Additionally, legibility is limited to simple patterns and text because of the wide spacing of the LED strips we use and the natural drop in edge pixel density caused by varying rotation radii.



(a) Arms retracted.

(b) Arms fully extended.

Figure 3: Demonstration of AeroScale prototype.

Limitations. While our prototype demonstrates feasibility at the proof-of-concept level, we acknowledge two critical challenges that must be addressed for future real-world deployment: (i) Further developing flight attitude control algorithms to enable (1) the drone to fly at an inclined angle, overcoming the current limitation of the display mainly facing downward and improving the viewing angle; and (2) enable stable flight and consistent display quality under the interplay among payload constraints, power supply limitations, mechanical forces from POV rotation, and flight control algorithms. (ii) Ensuring the operational safety, particularly regarding the risks posed by exposed rotating LED arms, potential crashes, and high-speed flying debris. To mitigate these risks, it is necessary to design a protective enclosure or safety cage that fully shields the rotating area, adopt high-toughness and fatigue-resistant materials (rather than 3D-printed plastics), and conduct structural stress analysis on the rotating assembly.

3 CONCLUSION AND FUTURE WORK

This paper introduced AeroScale, a drone-based, resizable mid-air UI designed to address the conflict between navigational mobility and shared visibility imposed by a fixed form factor. AeroScale achieves this by incorporating a retractable POV mechanism, enabling it to adapt its size to different environments and contexts, thereby offering a new degree of flexibility in designing mobile and shared mid-air AR applications.

For future work, we plan to extend this study in the following directions: (i) optimize the mechanical design by adopting lighter and more durable materials to develop multi-stage folding or umbrella-type structures, combined with a protective enclosure or safety cage to enhance visualization stability and overall safety; (ii) develop flight control algorithms that allow drones to maintain a stable attitude and flight path, accommodating both the system's variable size and diverse flight angles; (iii) conduct comprehensive evaluations, ranging from battery efficiency measurements to user studies investigating display legibility, perception, and social acceptance in real-world environments; and (iv) explore practical AR applications and services that leverage AeroScale's unique capabilities.

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